AI ASSISTANT CODING
ASSIGNMENT-11.5

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Batch: 34

Semester: VI

Branch: CSE(AIML)

Lab 11 – Data Structures with AI: Implementing Fundamental Structures

Task Description #1 – Stack Implementation

Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Sample Input Code:

class Stack:

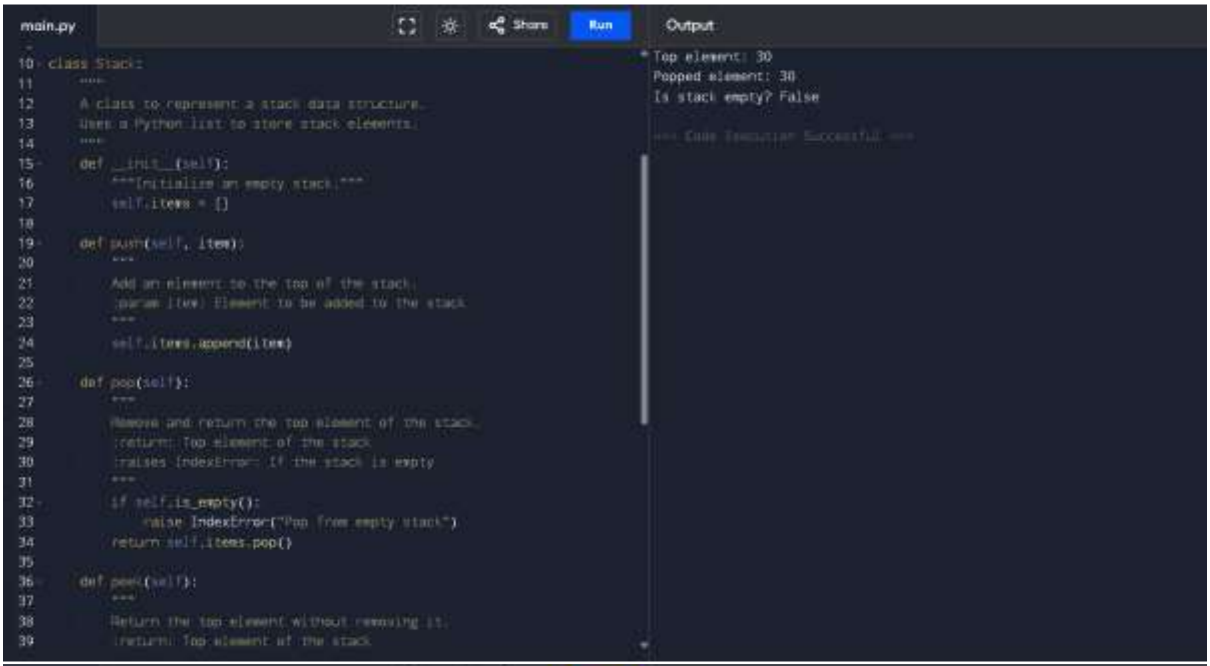
pass

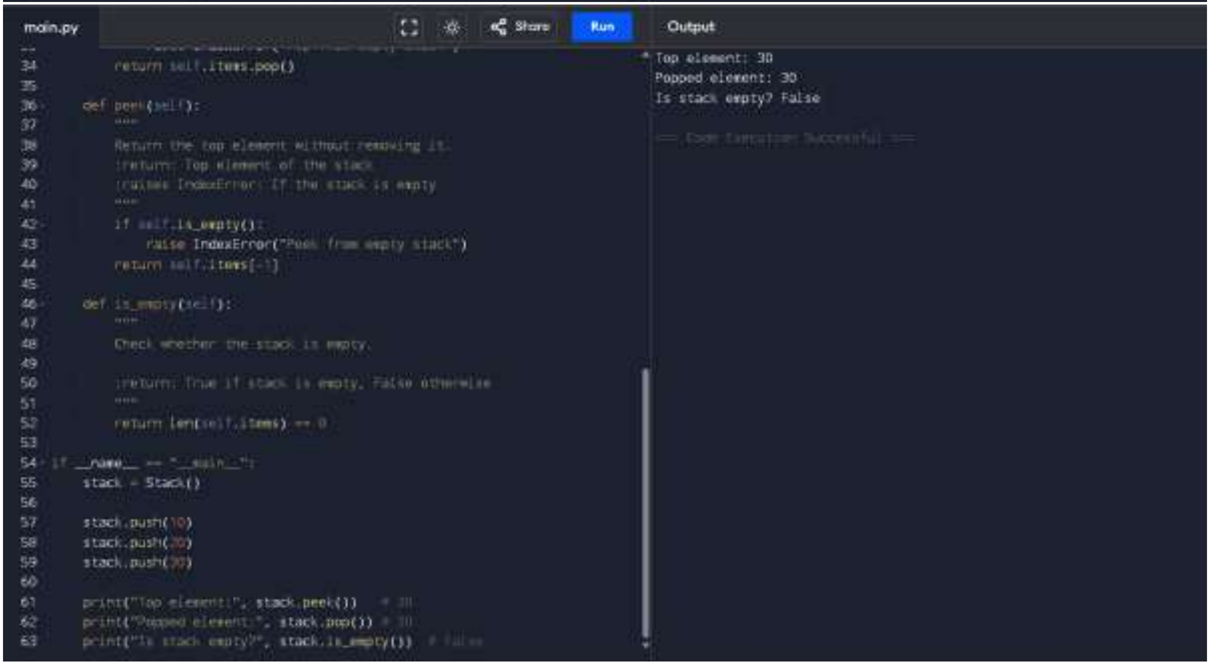
Expected Output:

• A functional stack implementation with all required methods and

docstrings.

```
"""Prompt:
Generate a Python class named Stack that implements the stack data
    structure using a list. The class should include the following
    methods:
push(item) – to add an element to the top of the stack
pop() – to remove and return the top element
peek() – to return the top element without removing it
is_empty() – to check whether the stack is empty
Include proper docstrings for the class and each method. Handle edge
    cases such as popping from an empty stack."""
```





Task Description #2 – Queue Implementation

Task: Use AI to implement a Queue using Python lists.

Sample Input Code:

class Queue:

pass

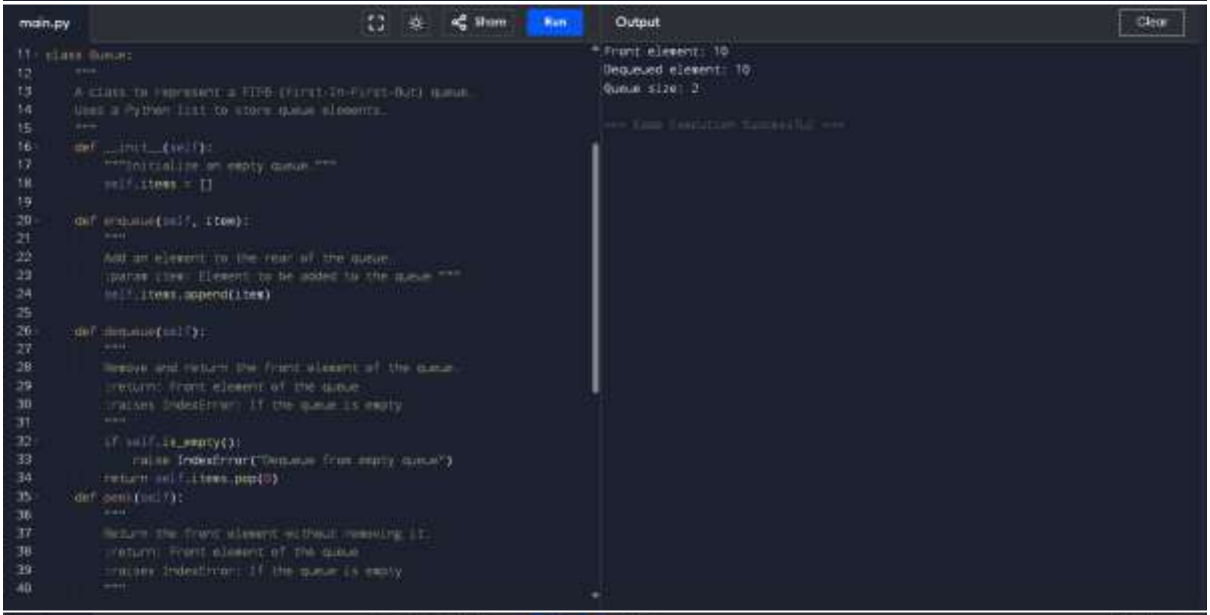
Expected Output:

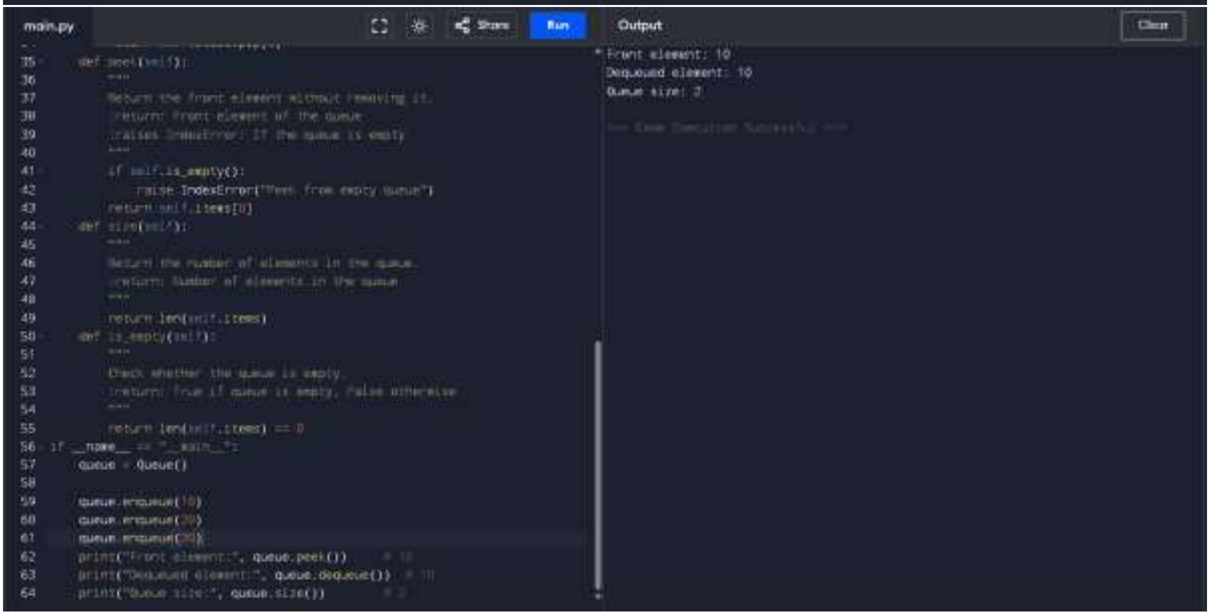
• FIFO-based queue class with enqueue, dequeue, peek, and size

methods.

```
""" Prompt:
Generate a Python class named Queue that implements a FIFO (First-In-First
-Out) queue using a Python list. The class should include the following
methods:

enqueue(item) – to add an element to the rear of the queue
dequeue() – to remove and return the front element
peek() – to return the front element without removing it
size() – to return the number of elements in the queue """
```





Task Description #3 – Linked List

Task: Use AI to generate a Singly Linked List with insert and displaymethods.

Sample Input Code:

class Node:

pass

class LinkedList:

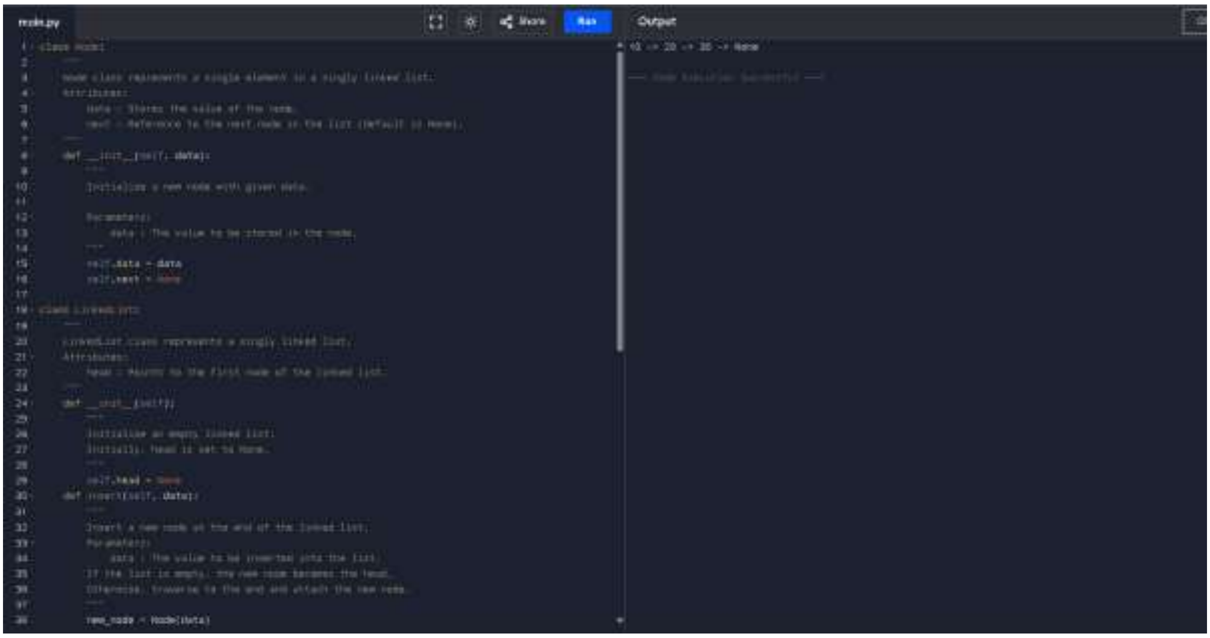
pass

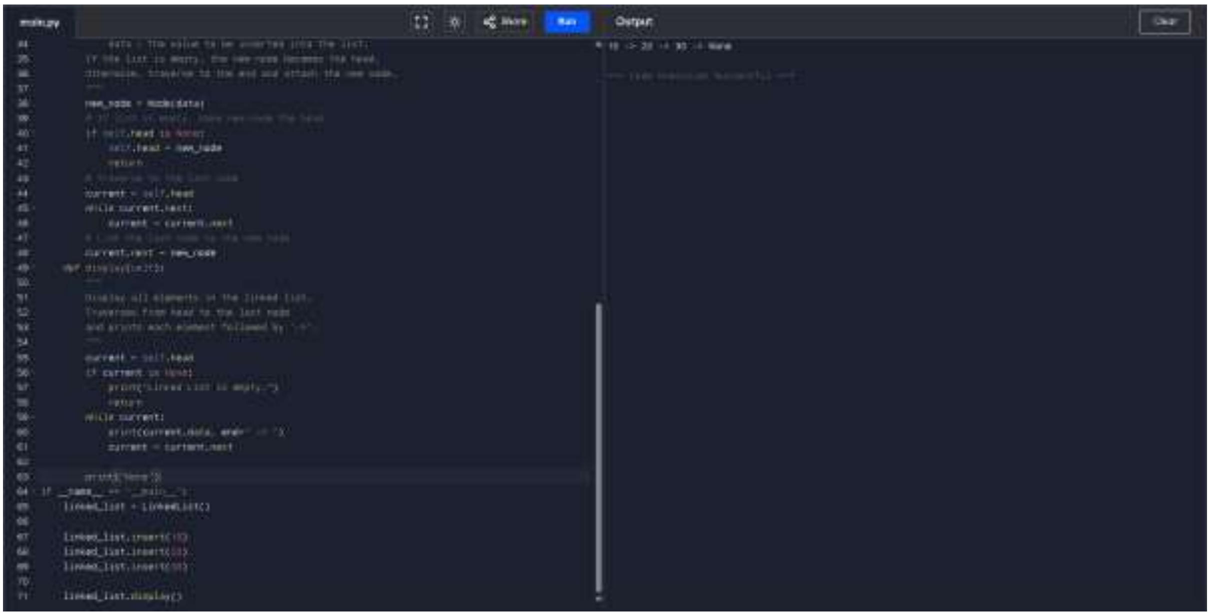
Expected Output:

• A working linked list implementation with clear method documentation

```
""" Prompt:
Generate a Python implementation of a Singly Linked List.
Create a Node class to store data and a reference to the next node.
Create a LinkedList class with the following methods:

insert(data) – to insert a new node at the end of the list
display() – to display all elements of the linked list
Include clear docstrings for all classes and methods. """
```





Task Description #4 – Hash Table

Task: Use AI to implement a hash table with basic insert, search, and delete methods.

Sample Input Code:

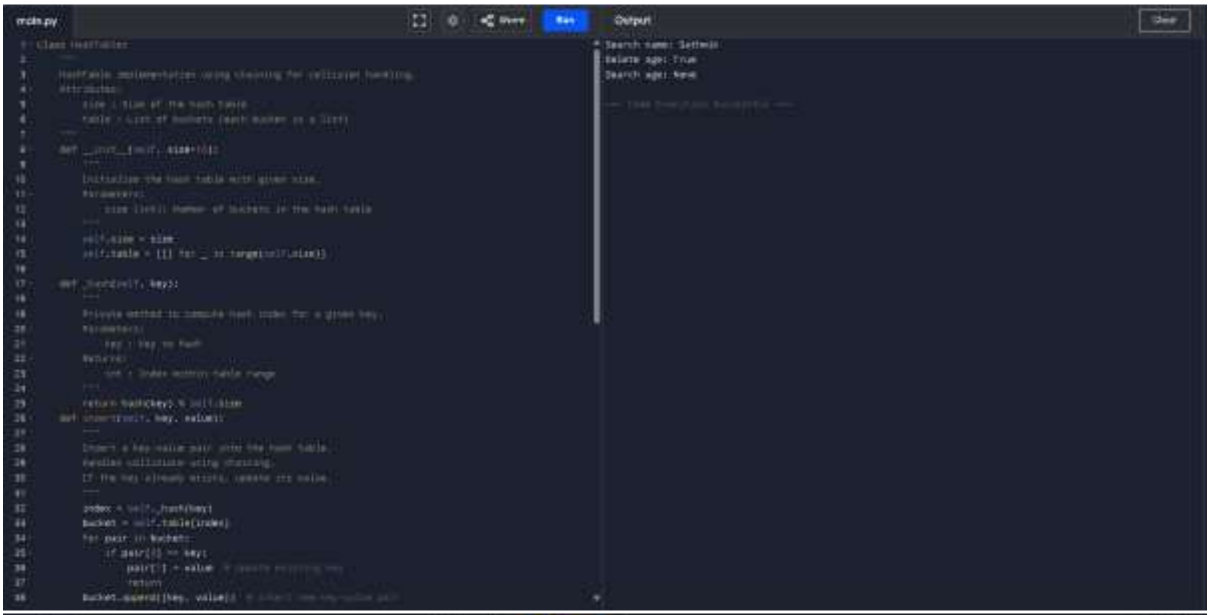
class HashTable:

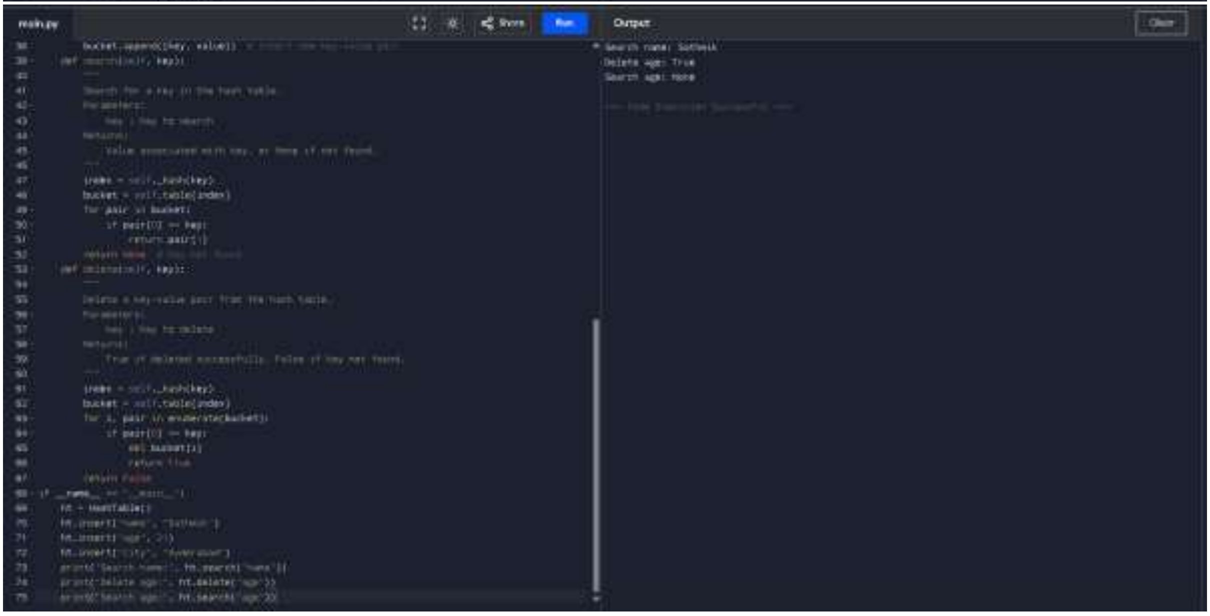
pass

Expected Output:

• Collision handling using chaining, with well-commented methods

```
"""Task Description #4 – Hash Table
Task: Use AI to implement a hash table with basic insert, search, and delete
methods.
Sample Input Code:
class HashTable:
pass
Expected Output:
• Collision handling using chaining, with well-commented methods. """
```





Task Description #5 – Graph Representation

Task: Use AI to implement a graph using an adjacency list.

Sample Input Code:

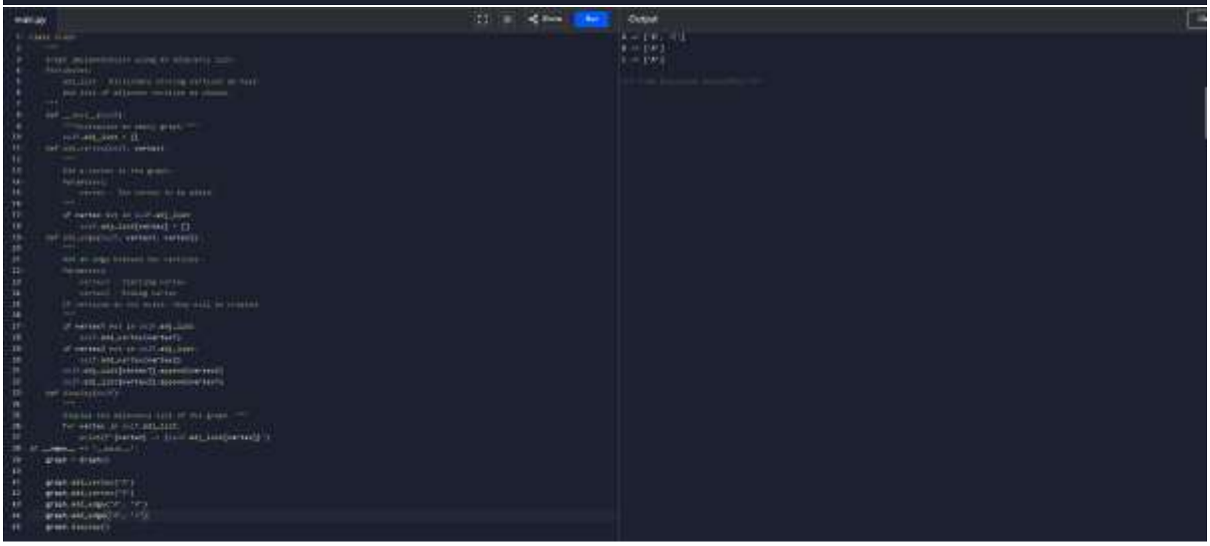
class Graph:

pass

Expected Output:

• Graph with methods to add vertices, add edges, and display connections.

```
""" Prompt:
Generate a Python class named Graph that represents a graph using an adjacency list.
The class should include the following methods:
add_vertex(vertex) - to add a new vertex to the graph
add_edge(vertex1, vertex2) - to add an edge between two vertices
display() - to display all vertices and their connected neighbors. """
```



Task Description #6: Smart Hospital Management System – Data Structure Selection

A hospital wants to develop a Smart Hospital Management System that handles:

1. Patient Check-In System – Patients are registered and treated in order of arrival.

2. Emergency Case Handling – Critical patients must be treated first.

3. Medical Records Storage – Fast retrieval of patient details using ID.

4. Doctor Appointment Scheduling – Appointments sorted by time.

5. Hospital Room Navigation – Represent connections between wards and rooms.

Student Task:

• For each feature, select the most appropriate data structure from the list below:

o Stack

o Queue

o Priority Queue

o Linked List

o Binary Search Tree (BST)

o Graph

o Hash Table

o Deque

• Justify your choice in 2–3 sentences per feature.

• Implement one selected feature as a working Python program with AI-assisted code generation.

Smart Hospital Management System – Data Structure Selection

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1. Patient Check-In System

Selected Data Structure: Queue

Justification:
A Queue follows the First-In-First-Out (FIFO) principle. Patients are treated in the order they arrive, ensuring fairness. Therefore, a Queue is the most suitable structure for managing regular patient check-ins.

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2. Emergency Case Handling

Selected Data Structure: Priority Queue

Justification:
Emergency patients must be treated based on severity rather than arrival time. A Priority Queue allows higher-priority (critical) patients to be treated first. This ensures immediate attention to life-threatening cases.

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3. Medical Records Storage

Selected Data Structure: Hash Table

Justification:
Patient records need fast retrieval using a unique patient ID. A Hash Table provides average O(1) time complexity for search, insert, and delete operations. This makes record management efficient and scalable.

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4. Doctor Appointment Scheduling

Selected Data Structure: Binary Search Tree (BST)

Justification:
Appointments must be maintained in sorted order based on time. A Binary Search Tree automatically keeps elements sorted. This makes scheduling and time-based retrieval efficient.

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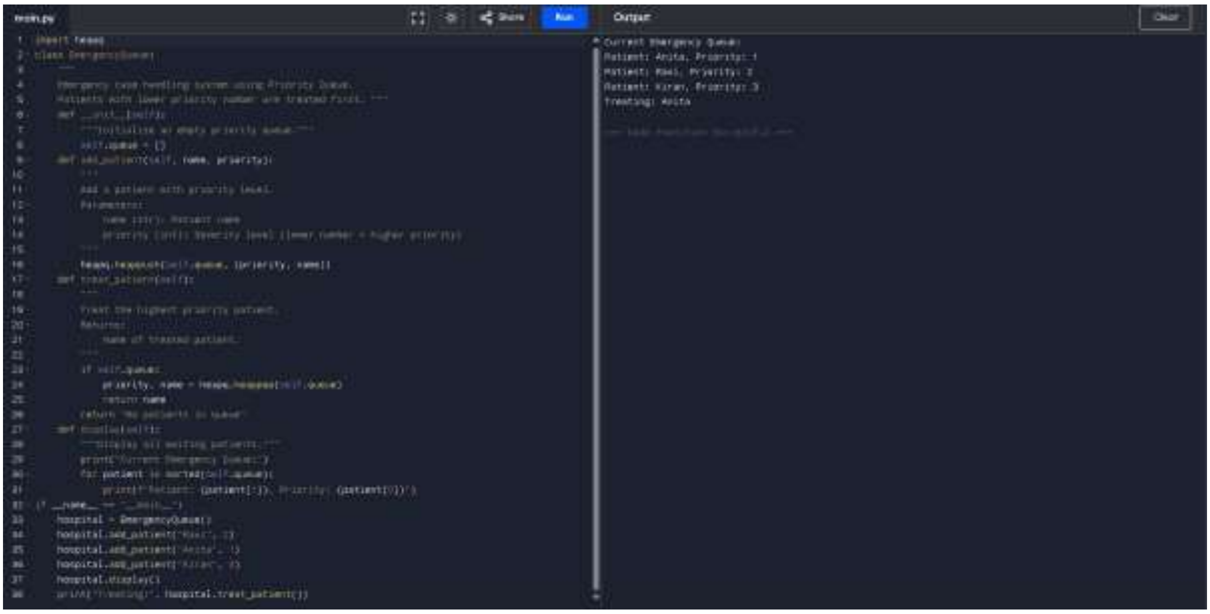
5. Hospital Room Navigation

Selected Data Structure: Graph

Justification:
Hospital rooms and wards can be represented as nodes, and pathways as edges. A Graph effectively models connections between locations. It also supports pathfinding for navigation within the hospital.



Task Description #7: Smart City Traffic Control System

A city plans a Smart Traffic Management System that includes:

1. Traffic Signal Queue – Vehicles waiting at signals.

2. Emergency Vehicle Priority Handling – Ambulances and fire trucks prioritized.

3. Vehicle Registration Lookup – Instant access to vehicle details.

4. Road Network Mapping – Roads and intersections connected logically.

5. Parking Slot Availability – Track available and occupied slots.

Student Task

• For each feature, select the most appropriate data structure from the list below:

o Stack

o Queue

o Priority Queue

o Linked List

o Binary Search Tree (BST)

o Graph

o Hash Table

o Deque

• Justify your choice in 2–3 sentences per feature.

• Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

• A table mapping feature → chosen data structure → justification.

• A functional Python program implementing the chosen feature with comments and docstrings.

Smart City Traffic Control System

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1. Traffic Signal Queue

Selected Data Structure: Queue

Justification:
Vehicles waiting at a traffic signal follow the First-In-First-Out (FIFO) principle. The vehicle that arrives first should move first when the signal turns green. Therefore, a Queue is the most appropriate structure for managing traffic signals fairly and efficiently.

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2. Emergency Vehicle Priority Handling

Selected Data Structure: Priority Queue

Justification:
Emergency vehicles such as ambulances and fire trucks must be given immediate access regardless of arrival time. A Priority Queue allows higher-priority vehicles to be processed before regular vehicles. This ensures faster emergency response and improved public safety.

---

3. Vehicle Registration Lookup

Selected Data Structure: Hash Table

Justification:
Vehicle details must be accessed instantly using a registration number. A Hash Table provides average O(1) time complexity for search operations. This makes it ideal for fast and efficient vehicle information retrieval.

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4. Road Network Mapping

Selected Data Structure: Graph

Justification:
Roads and intersections can be represented as nodes and edges. A Graph efficiently models connections between locations and supports route-finding algorithms. This makes it suitable for navigation and traffic management systems.

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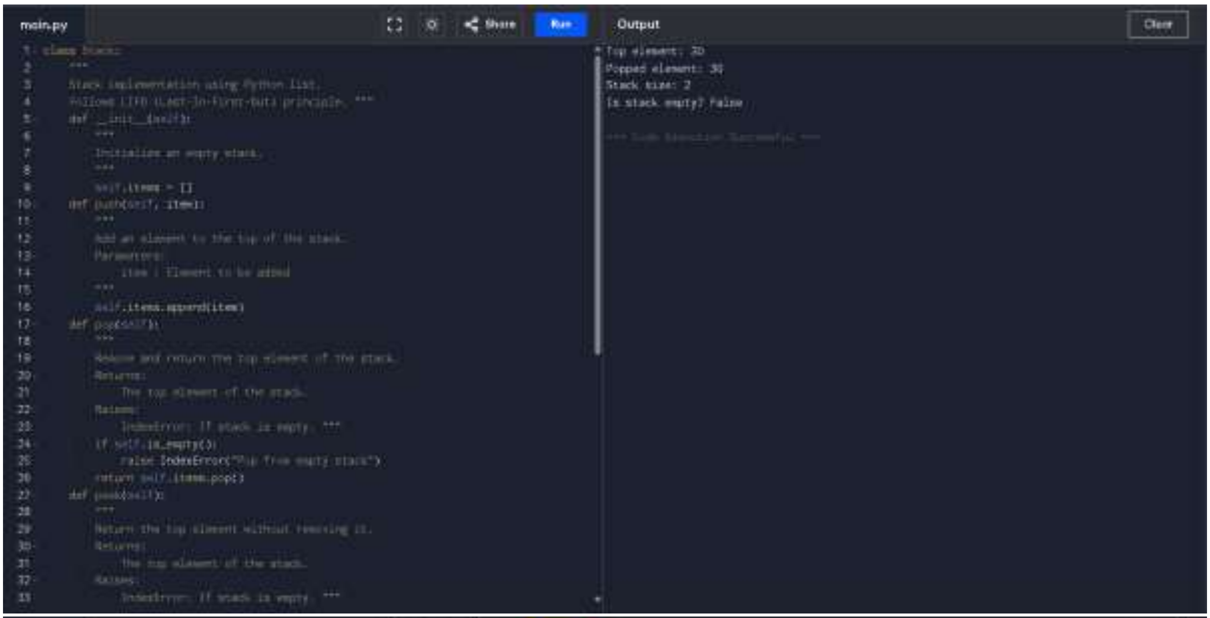
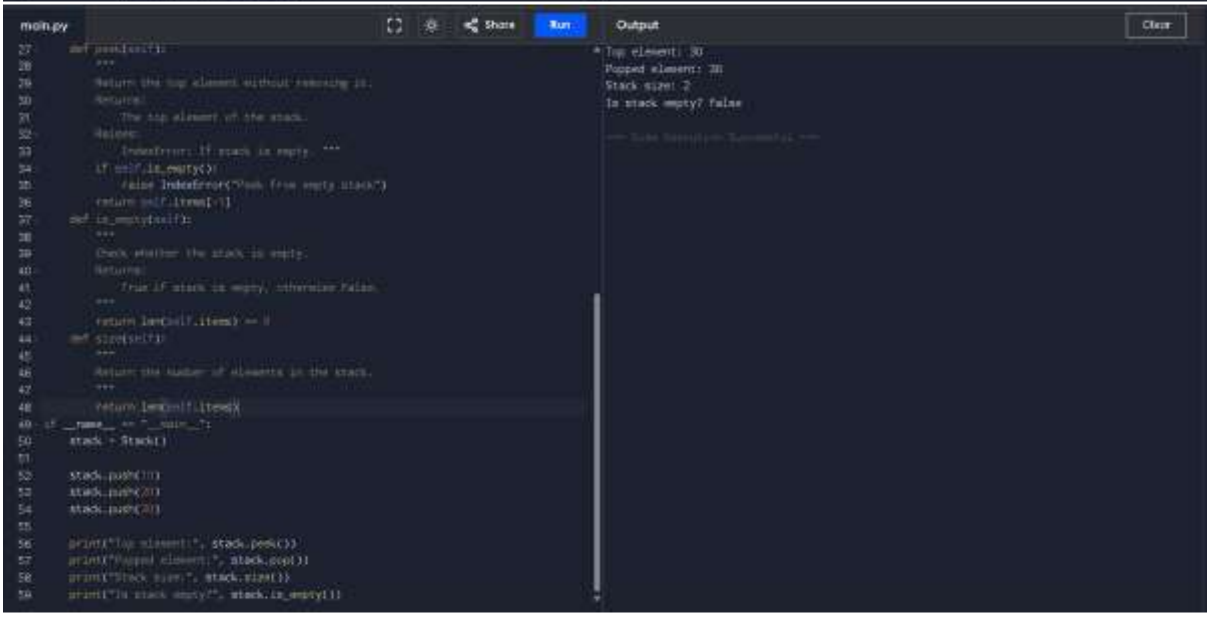
5. Parking Slot Availability

Selected Data Structure: Binary Search Tree (BST)

Justification:
Parking slots can be stored in sorted order based on slot numbers. A Binary Search Tree maintains elements in sorted order and allows efficient insertion, deletion, and searching. This makes it suitable for tracking available and occupied parking slots.





Task Description #8: Smart E-Commerce Platform – Data Structure.

Challenge

An e-commerce company wants to build a Smart Online Shopping System with:

1. Shopping Cart Management – Add and remove products dynamically.

2. Order Processing System – Orders processed in the order they are Placed.

3. Top-Selling Products Tracker – Products ranked by sales count.

4. Product Search Engine – Fast lookup of products using product ID.

5. Delivery Route Planning – Connect warehouses and delivery locations.

Student Task

• For each feature, select the most appropriate data structure from

the list below:

o Stack

o Queue

o Priority Queue

o Linked List

o Binary Search Tree (BST)

o Graph

o Hash Table

o Deque

• Justify your choice in 2–3 sentences per feature.

• Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

• A table mapping feature → chosen data structure → justification.

• A functional Python program implementing the chosen feature with comments and docstrings.

Smart E-Commerce Platform – Data Structure Selection

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1. Shopping Cart Management

Selected Data Structure: Linked List

Justification:
Products in a shopping cart are added and removed dynamically. A Linked List allows

efficient insertion and deletion without shifting elements. This makes it suitable for managing frequently changing cart items.

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2. Order Processing System

Selected Data Structure: Queue

Justification:
Orders must be processed in the order they are placed. A Queue follows the First-In-First-Out (FIFO) principle, ensuring fairness. Therefore, it is ideal for handling customer orders.

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3. Top-Selling Products Tracker

Selected Data Structure: Priority Queue

Justification:
Products need to be ranked based on sales count. A Priority Queue allows items with higher sales to be accessed first. This ensures efficient tracking of top-selling products.

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4. Product Search Engine

Selected Data Structure: Hash Table

Justification:
Products are searched using unique product IDs. A Hash Table provides average O(1) time complexity for search operations. This makes product lookup fast and efficient.

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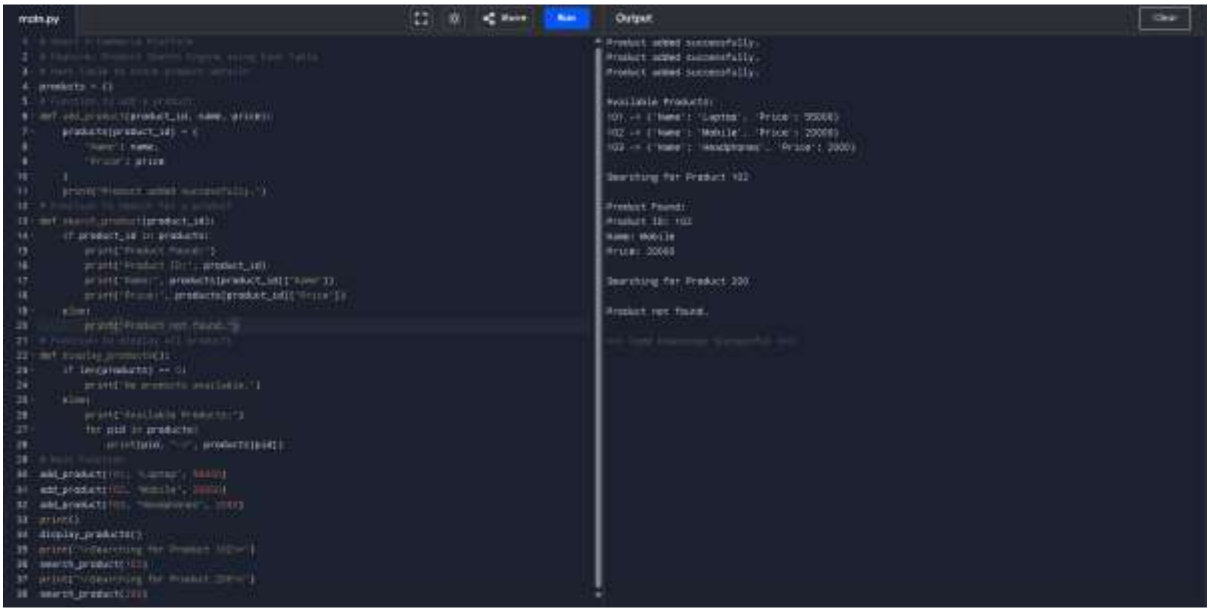
5. Delivery Route Planning

Selected Data Structure: Graph

Justification:
Warehouses and delivery locations can be represented as nodes, and roads as edges. A Graph efficiently models connections and supports shortest path algorithms. This makes it suitable for route optimization and logistics planning.


main.py
Run
Output
Clear
products = {}
products[product_id] = {
for pid in products:
display_products()
Product added successfully.
Product added successfully.
Product added successfully.
Available Products:
Searching for Product 102
Product Found:
Product ID: 102
Name: Mobile
Price: 20000
Searching for Product 200
Product not found.